CLASS 10 · CBSE · NCERT MATHEMATICS · FORMULA BANK

Chapter 1: Real Numbers

Every formula and result of the chapter on one sheet

1	Prime factorisation (Fundamental Theorem of Arithmetic)	$x = p_1^{a_1} \times p_2^{a_2} \times \cdots \times p_n^{a_n}$, unique for every composite number x
2	HCF by prime factorisation	Product of the smallest power of each common prime factor
3	LCM by prime factorisation	Product of the greatest power of every prime factor involved
4	HCF and LCM of two numbers	$\text{HCF}(a, b) \times \text{LCM}(a, b) = a \times b$
5	LCM from HCF	$\text{LCM}(a, b) = \dfrac{a \times b}{\text{HCF}(a, b)}$
6	HCF from LCM	$\text{HCF}(a, b) = \dfrac{a \times b}{\text{LCM}(a, b)}$
7	LCM of three numbers	$\text{LCM}(p, q, r) = \dfrac{p \cdot q \cdot r \cdot \text{HCF}(p, q, r)}{\text{HCF}(p, q) \cdot \text{HCF}(q, r) \cdot \text{HCF}(p, r)}$
8	HCF of three numbers	$\text{HCF}(p, q, r) = \dfrac{p \cdot q \cdot r \cdot \text{LCM}(p, q, r)}{\text{LCM}(p, q) \cdot \text{LCM}(q, r) \cdot \text{LCM}(p, r)}$
9	Co-prime numbers	$\text{HCF}(a, b) = 1$, so $\text{LCM}(a, b) = a \times b$
10	Number ending in 0	Its prime factorisation contains both 2 and 5
11	Prime dividing a square	If a prime p divides a^2, then p divides a
12	Rational and irrational	rational $\pm$ irrational = irrational; non-zero rational $\times$ irrational = irrational
13	Square root of a prime	$\sqrt{p}$ is irrational for every prime p